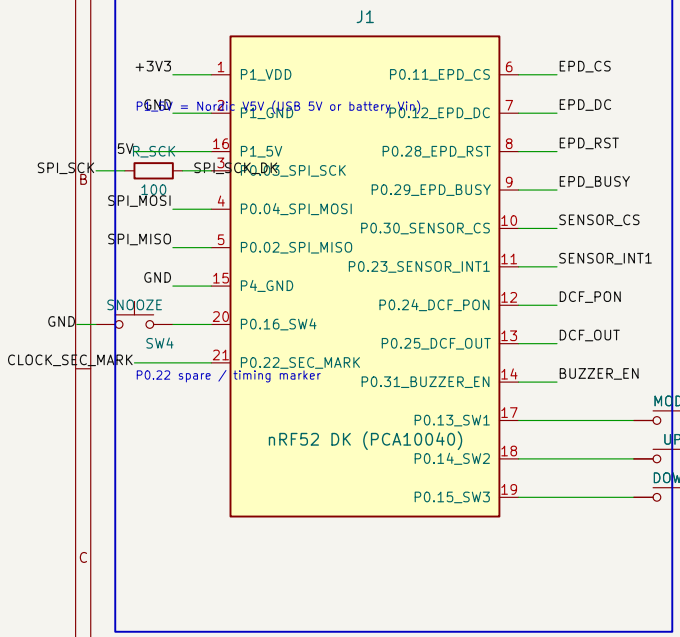
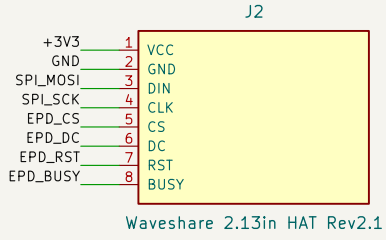
BLOCK 1: HOST CONTROLLER INTERFACE (nRF52 DK / PCA10040)

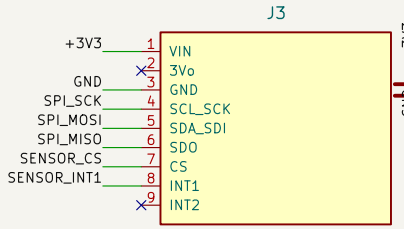


BLOCK 2: DISPLAY INTERFACE (Waveshare 2.13in HAT)

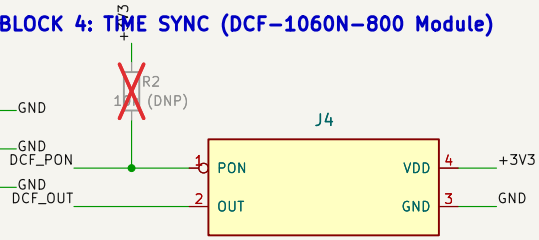


Verify BS1 = 0 (4-wire SPI) on HAT rear switch

BLOCK 3: MOTION SENSOR (Adafruit LIS3DH #2809)



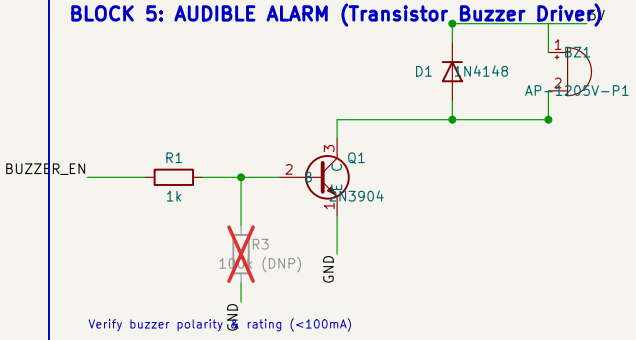
BLOCK 4: TIME SYNC (DCF-1060N-800 Module)



PON = 0V: Enabled (~68 uA) | PON = 3.3V: Disabled (0.0 uA)

MEASURE OUT HIGH LEVEL BEFORE CONNECTING DK P0.25!

BLOCK 5: AUDIBLE ALARM (Transistor Buzzer Driver)

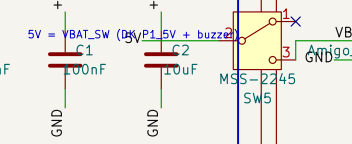


Verify buzzer polarity & rating (<100mA)

BLOCK 6: BATTERY IN (plain LiPo Amigo PIM

Amigo BAT/VBAT → J5; SW5 MSS-2245 one pole (comm

Prefer switch OFF while charging (USB-C on Amigo onl



PROTOTYPE SAFETY & INTEGRATION RULES

1. POWER MODES: (A) USB bring-up: 3V3 from P1 VDD only; do NOT attach battery to P1 5V. (B) Battery demo: USB unplugged; Akyga→plain LiPo Amigo BAT/VBAT→MSS-2245 (SW5)→P1 5V + buzzer+. Amigo 5V pads = USB VBUS only (no SYS/DEVICE jack on PIM611). NEVER dual-feed USB+batter

2. SAFE CONTROL STATES: Set DCF_PON HIGH (disabled, 0.0uA). Set EPD_CS and SENSOR_CS HIGH (unselected). Set BUZZER_EN LOW (off).

3. DCF-77 VERIFICATION: Measure DCF OUT high-level voltage before connecting to DK P0.25. Orient antenna perpendicular to Frankfurt.

4. E-PAPER CONFIGURATION: Verify switch BS1 = 0 on back of Waveshare HAT for 4-wire SPI mode before powering display rail.

5. BUZZER DRIVER: Verify 2N3904 TO-92 pinout (E-B-C). Buzzer+ and D1 cathode on 5V/VBAT_SW — NOT 3V3 (buzzer is silent on 3V3).

6. STAGED BRING-UP: 1) DK Blinky -> 2) E-Paper Demo -> 3) LIS3DH WHO_AM_I -> 4) DCF Standalone -> 5) DCF Sync -> 6) Buzzer Alarm.

SHIELD v0.3 POWER: plain LiPo Amigo (PIM611) is off-board charger. Akyga on Amigo JST. Bring Amigo BAT/VBAT + GND onto shield J5. SW5 = MSS-2245 (DPDT physical: one pole drawn as SPDT) in series VBAT→5V (=VBAT_SW→DK P1_5V + BZ1). Leave Fast Charge pads open. Socket footprints st

nRF52 DK shield: R_SCK, buttons, Amigo VBAT→MSS-2245→5V/buzzer

Bachelor Thesis (APD)

Sheet: /

File: dk-perfboard-shield.kicad_sch

Title: Ultra-Low-Power Bedside Clock — DK Perfboard Shield

Size: A4	Date: 2026-11-03	**Rev: v1.0-shield**
KiCad E.D.A. 10.0.4		Id: 1/1